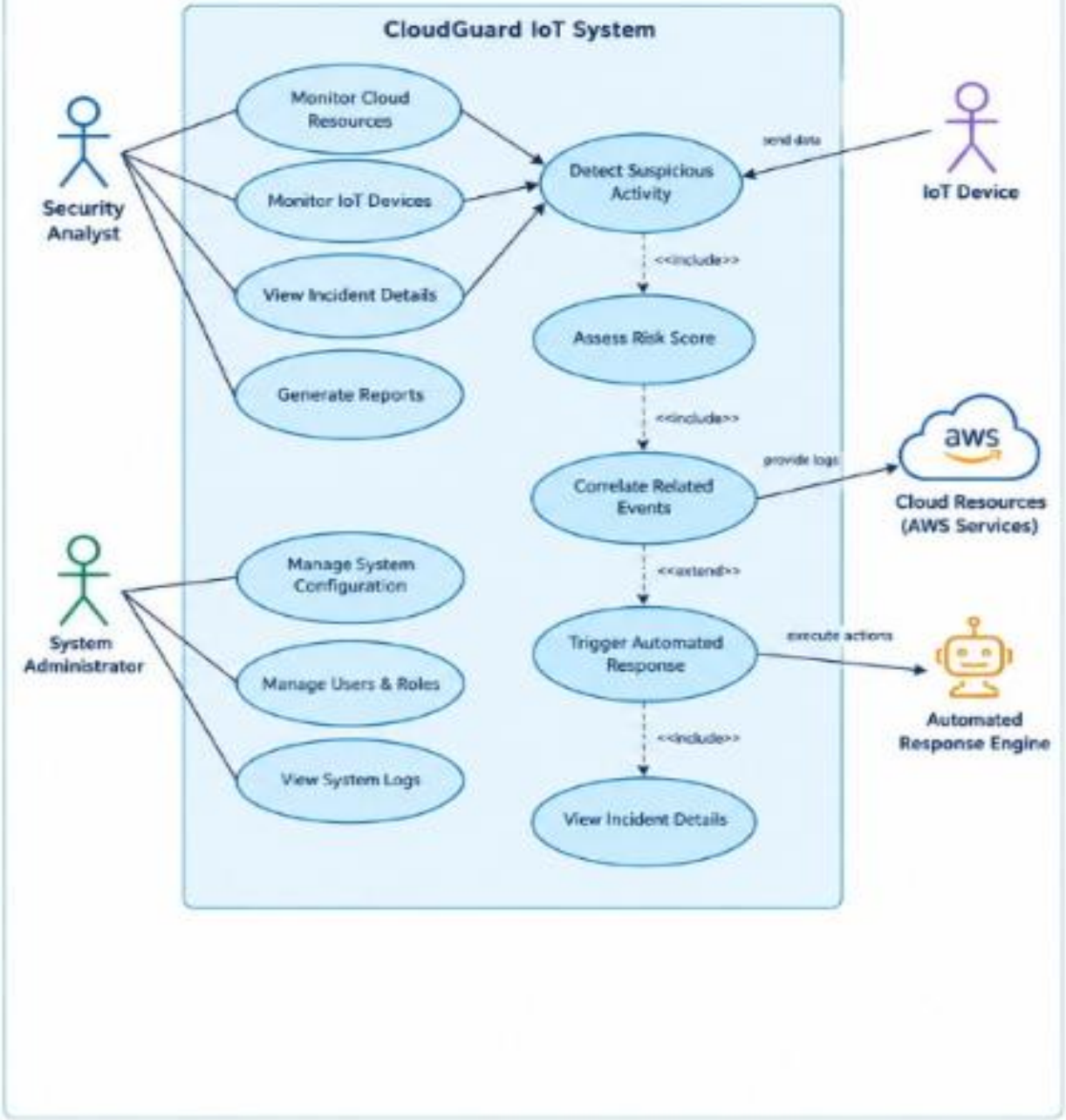
# CloudGuard IoT

Unified Autonomous Cloud & IoT Security Platform

## Use Case Diagram



# CloudGuard IoT — Use Cases

## 1. Overview

This document describes the main use cases of the CloudGuard IoT system. The use cases illustrate how different actors interact with the system to monitor, detect, analyze, and respond to security events involving cloud resources and IoT devices.

## 2. Actors

| Actor | Description |
| --- | --- |
| Security Analyst | Monitors alerts, investigates incidents, and manages responses. |
| System Administrator | Manages system configuration, users, roles, and infrastructure settings. |
| IoT Device | Sends telemetry data and status information to AWS IoT Core. |
| Cloud Resources | AWS services, such as EC2, S3, and Lambda. |
| Automated Response Engine | Executes predefined security actions, such as isolating devices or restricting affected resources. |

## 3. Use Cases

### UC-01: Monitor Cloud Resources

**Primary Actor: Security Analyst**

Description: Allows the security analyst to review cloud activity, logs, and available security findings from selected AWS services.

Preconditions:

The required cloud event sources are configured.

The system has permission to access the selected monitoring data.

Main Flow:

1. The system collects or receives cloud security events.
2. The system processes the available event data.
3. The security analyst opens the cloud monitoring dashboard.
4. The system displays the available findings and resource information.

Postcondition:

Cloud monitoring information is available for review.

UC-02: Monitor IoT Devices

Primary Actor: Security Analyst

Description: Tracks registered IoT devices, their telemetry, status, and selected behavioral indicators.

Preconditions:

IoT devices are registered or represented in the test environment.

The relevant telemetry or event source is configured.

Main Flow:

1. An IoT device sends telemetry or status information.
2. The system receives or retrieves the available device data.
3. The system evaluates the device's recorded behavior.
4. The dashboard displays device status and relevant observations.

Postcondition:

The analyst can review the recorded status and behavior of monitored IoT devices.

UC-03: Detect Suspicious Activity

Primary Actor: System

Description: Identifies selected suspicious patterns in cloud and IoT events using configured detection rules.

Preconditions:

Event sources and detection rules are configured.

Relevant event data is available.

Main Flow:

1. The system receives an event.

2. The detection component evaluates the event against configured rules.

3. If a rule matches, the system creates a security finding.

4. The finding is recorded with available event details and timestamps.

Postcondition:

A detected event is recorded for risk assessment and further investigation.

UC-04: Assess Risk Score

Primary Actor: System

Description:  Calculates an explainable risk score based on predefined indicators and their configured weights.

Preconditions:

* A security finding or relevant event exists.

* Risk indicators, weights, and severity thresholds are configured.

Main Flow:

1. The system retrieves the relevant event details.

2. The risk engine evaluates applicable indicators.

3. The engine calculates a risk score.

4. The system assigns a severity level according to the configured thresholds.

5. The system records the score and its contributing factors.

Postcondition:

The finding has a risk score and an explainable severity classification.

UC-05: Correlate Related Events

Primary Actor: System

Description: Associates related cloud and IoT events with a unified incident when they satisfy predefined correlation criteria.

Preconditions:

Relevant events have been detected or received.

Correlation rules are configured.

Main Flow:

1. The system evaluates event attributes, timestamps, and entity identifiers.
2. The correlation engine checks whether the events meet the configured criteria.
3. Matching events are associated with an existing incident or a new incident is created.
4. The system records the associated events and correlation rationale.
5. The incident risk information is updated when appropriate.

Postcondition:

Related events are linked to an incident when the configured criteria are satisfied.

UC-06: Trigger Automated Response

Primary Actor:  Automated Response Engine

Description: Executes a predefined containment workflow when an incident satisfies the configured response conditions.

Preconditions:

An incident meets the configured response threshold.

Automated response is enabled.

The target device or resource is explicitly authorized for the action.

Main Flow:

1. The system evaluates the incident's risk and response conditions.
2. The response component verifies that the target is within the approved scope.
3. The system invokes the configured containment action.
4. The response outcome is recorded.
5. The incident and entity status are updated when the outcome can be confirmed.

Postcondition:

The response attempt and its outcome are recorded for review.

Safety Note: Automated containment must be restricted to designated test devices and cloud resources during project development and demonstration.

UC-07: View Incident Details

Primary Actor:  Security Analyst

Description: Allows the analyst to investigate an incident by reviewing its severity, related events, affected entities, and response history.

Preconditions:

An incident record exists.

The analyst has permission to access the incident details.

Main Flow:

1. The analyst opens the incidents dashboard.
2. The analyst selects an incident.
3. The system displays its details, risk factors, and related events.
4. The system presents available response records and evidence references.
5. The analyst reviews the incident timeline and status.

Postcondition:

The analyst can understand the recorded incident and its handling history.

UC-08: Manage System Configuration

Primary Actor: System Administrator

Description: Allows an authorized administrator to manage supported system settings, detection rules, risk thresholds, and response configuration.

Preconditions:

The administrator is authenticated and authorized.

Configuration management functionality is available.

Main Flow:

1. The administrator opens the configuration interface.
2. The system displays the settings the administrator is permitted to manage.
3. The administrator updates an approved setting.
4. The system validates the change.
5. The system saves the change and records an audit entry where supported.

Postcondition: The configuration is updated or the change is rejected with an appropriate error.

UC-09: Generate Reports

Primary Actor: Security Analyst

Description: Allows the analyst to review historical incidents and produce reports based on available security and response records.

Preconditions:

Incident or event records are available.

The analyst has permission to access the records.

Main Flow:

1. The analyst opens the reporting view.
2. The analyst selects the available reporting criteria.
3. The system retrieves the matching records.
4. The system presents a summary of findings, incidents, and response outcomes.
5. The analyst exports the report if export functionality is implemented.

Postcondition: A report is displayed or generated from the available records.

## 4. Relationships Between Use Cases

UC-03 Detect Suspicious Activity : provides findings for - UC-04 Assess Risk Score.

UC-04 Assess Risk Score : provides risk information used by - UC-05 Correlate Related Events and response decision logic.

UC-05 Correlate Related Events : may create or update incidents reviewed through UC-07 View Incident Details.

UC-06 Trigger Automated Response : may be initiated when an incident satisfies the configured response conditions.

UC-08 Manage System Configuration : maintains supported rules and settings used by monitoring, risk assessment, and response components.

UC-09 Generate Reports : summarizes available event, incident, and response records.

These relationships describe the intended workflow. In the UML diagram, use <<include>> only for behavior that is required as part of another use case, and <<extend>> only for optional or conditionally executed behavior.

## 5. Additional Notes

The system is designed to monitor selected cloud resources and IoT devices through a unified security workflow.

Risk scores are prioritization indicators; they do not independently prove that an attack has occurred.

Event correlation depends on the quality of available data and the configured correlation rules.

Automated actions must be authorized, logged, and tested in a controlled environment.

The final implementation may support only a subset of the use cases, depending on AWS permissions, service availability, cost, and project time.

Each use case should be linked to relevant functional requirements and test cases before final submission.